

Child Survival and the Decline in Hours Worked

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Abstract

This paper develops a model to depict the interplay between child mortality, fertility and working hours. In it, parents partially control their children's chances of survival through their expenditure on food, medicines, etc. Due to this assumption, having a child increases the household's demand for goods, and thus, the supply of labour along the intensive margin. By contrast, improvements in child mortality lead parents to raise fewer children and invest more in their education. The combination of these two mechanisms implies then that demographic transitions come along with trends in the use of time as those observed since the Industrial Revolution.

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1 Introduction

It is well known that all developed societies have experienced major economic and demographic changes since the Industrial Revolution. For example, fertility rates have followed a path of secular decline that continues to this day. Likewise, working hours have steadily decreased to historically low levels. But although fertility and labour supply decisions are made jointly in the context of the family, their trends have been studied separately as independent phenomena until now. This paper contributes to the literature by filling the gap between demographic transitions and trends in the use of time, establishing a unified explanation for both.

So far, the literature provides three possible explanations for the decline in working hours. The first one is that preferences are such that wage growth induces the desire to enjoy more leisure time because the income effect is stronger than the substitution effect (see Boppart and Krusell (2020)). Another proposed hypothesis is that more extensive tax and transfer system discourage the supply of labour along the intensive margin (see McDaniel (2011) and Bick and Fuchs-Schündeln (2018)). More recently, it has been argued that reducing the cost of finding a job may result in fewer hours worked per worker (see Bick et al. (2022)).

Although the mentioned theories seem plausible explanations for why people work fewer hours and enjoy more leisure time, they only offer a partial or incomplete vision of the underlying mechanisms driving trends in the use of time. The reason is that they do not take into account the fact that fertility and labour supply decisions are intertwined. As a result, these theories do not explain why fertility rates decline at the same time as working hours do. Moreover, they also fail to explain the upwards trend in parents' amount of time devoted to educational childcare time (see Gauthier et al. (2004), Guryan et al. (2008) Gimenez-Nadal and Sevilla (2012), and Doepke et al. (2019)).

The hypothesis motivating this paper is that the observed long-run trends in the use of time are partly caused by changes to the demography. This hypothesis builds upon two empirically plausible mechanisms. The first is that improvements in child mortality rates encourage parents to raise fewer children (see Herzer et al. (2012), Wilson (2015), and Ager et al. (2018)). The second mechanism is that parents with more children tend to work more hours to finance a higher consumption of goods such as food, clothes, medicines, etc. In other words, larger families supply more hours of work because their demand for goods is higher in comparison to smaller families with the same potential income. The interplay between these two mechanisms implies then that changes to the demography should affect the use of time, causing trends as observed in data.

In order to illustrate the interplay between child mortality, fertility, and labour

supply decisions, I develop a two periods overlapping-generations (OLG) in which time can be devoted to work, raising children, and educating them. In the model, parents are aware that spending more income per child improves their offspring's survival rate due to better nutrition, clothing, and health investment. These two assumptions imply that parents internalize the impact of their time allocation on both their children's human capital and chances of survival.

Essentially, the model studied here is similar to the one propounded by Strulik (2004b), whereby children's chances of survival depend on controllable (internal) and uncontrollable (external) factors. The main difference is that the controllable mortality elements such as nutrition, clothing, or health investment are improved by spending more income per child (consumption) instead of time (spending share). This difference turns out to be non-trivial because children's health becomes an increasing function of the number of hours worked through child consumption, rather than being a decreasing function of the time devoted to work.

The first main novelty of the model is the prediction of a positive gradient between working hours and fertility rates. Unlike previous models, this gradient emerges because the demand for goods depends on the number of members in the family, which is determined by parents' fertility choice. The consequence is that changes to the demography translate to the use of time in an empirically plausible way. Moreover, given that this gradient operates through child consumption, it holds even if leisure time or saving rates are endogenized as control variables.

The second main novelty of the model is what we call "internal or controllable child survival effect". As parents with higher hourly wages spend more income per child, their children have better chances of survival due to better nutrition, clothing, healthcare, etc. Better health in turn encourages parents to have fewer children and invest more in their education. Accordingly, parents with more human capital (higher hourly wages) allocate their time differently, investing more time in their children's education as the evidence points out (see Guryan et al. (2008), and Doepke et al. (2019)).

Similarly to Strulik (2004b), this model also predicts the existence of an "external child survival effect". The latter notion refers to changes in the optimal household allocations caused by variations in external or uncontrollable mortality factors. The novelty in this model is that the external survival effect also alters parents' labour supply decisions for the same reason as the internal effect. However, this model predicts that the external survival effect is weaker for wealthier parents because their children already enjoy a better health.

One of the main predictions of the model is thus that long-run economic growth can induce demographic transitions accompanied by changes in the use of time

consistent with the historical evidence. Moreover, given that the effects described above operate through internal and external mortality factors, the model can easily be extended to study the role of taxes and public spending. In summary, the model provides a description of the interplay between child mortality, fertility, and labour supply decisions that can explain their co-movement.

The remainder of the paper is organised as follows. The next section describes the model, while its predictions are presented in the third section. The last section makes some final considerations.

2 The Model

Time is discrete and periods are ordered by $t \in \mathbb{N}_+$. During each period, the economy is comprised by two overlapped generations called adults and children. Families are mono-parental, with all decisions being made by the sole parent in the household, while children remain passive. During adulthood, individuals chose to have b_t children, from which $0 < s_t < 1$ reach adulthood. For the sake of simplicity, we assume that children are homogeneous within households, so their parents treat them identically. Individuals may, nonetheless, differ in their human capital endowment h_t if they come from different families.

There are two features that differentiate this model from existing ones. The first difference is that parents distinguish between their own consumption and that of their children, denoted c_t and z_t respectively. Yet even more importantly, parents are aware that spending more income on consumption per child improves their children's chances of survival, that is, $\partial s / \partial z > 0$. This last assumption reflects the positive effect on children's health of better nutrition, clothing, healthcare, etc.

Parents have preferences defined over their own consumption c_t and the quantity-quality mix of their surviving children $s_t b_t h_{t+1}$, where h_{t+1} stands for children's quality or human capital. These preferences are represented by a standard logarithmic utility function of the form:

$$U[c, s, b, h] = \alpha \ln c_t + (1 - \alpha) \ln(s_t b_t h_{t+1}), \quad 1 > \alpha > 0. \quad (1)$$

Since child consumption z_t already provides utility through higher child survival rates, the inclusion of an explicit term for it in the utility function simply adds redundant parameters.

Regarding the budget constraint, the sole source of income is labour $w_t h_t n_t$, which depends on the adult's human capital h_t , the number of hours worked n_t , and the wage rate per efficient hour paid in the labour market w_t . Income is spent entirely purchasing units of a homogeneous final good that is consumed by the adult and

their children. There is no savings market, so the household's budget constraint reads as

$$w_t h_t n_t = c_t + b_t z_t. \quad (2)$$

Besides supplying labour, parents can devote time to childcare. According to the standard, raising a child involves a fixed time cost of $\bar{\epsilon}$ hours required for basic childcare. In addition, parents can invest ϵ_t hours per child to promote their quality. Denoting the total number of available hours by $T > 0$, the distribution of time between working and childcare must satisfy the time constraint

$$T = n_t + (\bar{\epsilon} + \epsilon_t) b_t. \quad (3)$$

Children's quality or human capital h_{t+1} depends on how much time their parents invest in their education ϵ_t according to the technology

$$h_{t+1} = (\theta + \epsilon_t) h_t. \quad (4)$$

Parent's human capital h_t enters the education technology as a multiplicative term to allow for sustained accumulation over generations. Parameter $\theta > 0$ measures the amount of human capital acquired through other channels, such as observation, genetics, social interactions or public school if there were. This education technology is used in the literature to stress that investing time in children's education is not an essential input in the process of skill formation (i.e., $h_{t+1} > 0$ even if $\epsilon_t = 0$). Nevertheless, parents may still invest time in their children's education because it fosters the process, which occurs if it holds that $\theta > \bar{\epsilon}$.²

Following Strulik (2004a,b), child survival rates are the composite function of controllable (internal) and uncontrollable (external) factors. Controllable elements such as the quality of nutrition or clothing are included in child consumption z_t . Uncontrollable factors such as the climate or the prevalence of diseases are in turn captured by the dynamic index X_t . For instance, public investment in the potable water infrastructure can be represented by a higher X_t . Formally, our assumptions about child mortality are represented by a survival function,

$$s_t = S[z_t, X_t], \quad (5)$$

which is twice-continuously differentiable, strictly increasing and concave in z_t and X_t . As argued by Strulik (2004a,b), improvements in the external mortality elements should reduce the effectiveness of the internal factors, leading to the assumption that

$$\frac{\partial^2 S[z_t, X_t]}{\partial z_t \partial X_t} \leq 0.$$

²This parametric condition implies that the fixed time cost of fertility $\bar{\epsilon}$ is not so high that investing more time in educating children is always suboptimal.

Under these hypotheses about the survival function, it is easy to verify that the spending elasticity of child survival

$$\Phi_t = \frac{\partial S[z_t, X_t]}{\partial z_t} \frac{z_t}{S[z_t, X_t]},$$

is a strictly decreasing function of both child consumption z_t and the dynamic index X_t . Finally, given that the risk of mortality can never equal zero, we impose the following Inada conditions:

$$\lim_{z \rightarrow \infty} \frac{\partial S[z_t, X_t]}{\partial z_t} = 0, \quad \lim_{X \rightarrow \infty} \frac{\partial S[z_t, X_t]}{\partial X_t} = 0.$$

Definition 1. *Given $\{h_t, w_t, X_t\}$, an allocation $\Omega_t^* \equiv \{c_t^*, b_t^*, \epsilon_t^*, z_t^*\}$ is said to be optimal if it maximises utility (1), subject to the budget constraint (2), the time restriction (3), the education technology (4) and the child survival function (5).*

When imposing non-negativity conditions on the solution, an optimal household allocation must satisfy the following system of first order conditions (FOCs):

$$\frac{\partial U}{\partial c} = \lambda_t, \tag{6}$$

$$\frac{\partial U}{\partial b} = \frac{\partial U}{\partial c} (z_t + w_t h_t (\epsilon_t + \bar{\epsilon})), \tag{7}$$

$$\left(\frac{\partial U}{\partial s} \frac{\partial s}{\partial z} - \frac{\partial U}{\partial c} b_t \right) z_t = 0, \tag{8}$$

$$\left(\frac{\partial U}{\partial h} \frac{\partial h}{\partial \epsilon} - \frac{\partial U}{\partial c} b_t w_t h_t \right) \epsilon_t = 0. \tag{9}$$

According to equation (6), the optimal adult consumption level c_t^* must be such that the marginal utility of consumption equals the marginal utility of income (shadow value) λ_t . This means that the cost of all the other choices is measured in terms of forgone parental consumption. For instance, condition (7) establishes that the marginal utility of having a child must equate its marginal cost in terms of forgone adult consumption. Similarly, if parents decide to spend income on their children, the optimal child consumption level must be such that its marginal utility equals the marginal cost involved (see equation (8)). Finally, condition (9) states that if parents decide to invest time in their children's education, its marginal utility and marginal cost must be the same.

Note that equations (6)-(8) uphold the following: parents always decide to have children $b_t^* > 0$, parent's consumption is positive $c_t^* > 0$, and children receive

a positive amount of goods $z_t^* > 0$. Otherwise, no children would survive and raising them would therefore be pointless. By contrast, equation (9) yields an interior solution, namely $\epsilon_t^* > 0$, only when the optimal fertility choice is below a certain threshold (see the Appendix). If this threshold is interpreted as the biological maximum, the optimal solution is always interior. In any case, we will abstract from the corner solution and focus on the behaviour of the interior one with positive time investment in children's education $\epsilon_t^* > 0$.³

A shortcoming of working with a general functional form for the child survival function (5) is that we cannot explicitly solve the system of FOCs (6)-(9). What can be done, however, is express the optimal household allocations as functions of the elasticity measure Φ_t^* and the optimal fertility rate b_t^* as follows:

$$c_t^* = w_t h_t \alpha T \quad (10)$$

$$z_t^* = w_t h_t (\theta - \bar{\epsilon}), \quad (11)$$

$$b_t^* = \frac{(1 - \alpha) T \Phi_t^*}{(\theta - \bar{\epsilon})}, \quad (12)$$

$$\epsilon_t^* = \frac{(1 - \alpha) T}{b_t^*} - \theta, \quad (13)$$

$$n_t^* = \alpha T + (\theta - \bar{\epsilon}) b_t^*. \quad (14)$$

Equations (10) and (11) show that every individual in the household consumes a constant fraction of the hourly wage. Recall that the elasticity measure Φ_t^* depends inversely on child consumption z_t^* (internal factors) and the dynamic index X_t (external factors). Equation (12) thus depicts the effect of internal and external survival factors on child quantity b_t^* . Finally, equations (13) and (14) describe how these effects impact upon the allocation of time through fertility.

3 Results

This model's main novelty is the existence of a positive gradient between working hours and fertility rates. The reason it emerges and the intuitions behind it will be discussed in the next subsection. The second novelty refers to what we call internal or controllable child survival effect, as opposed to the external or uncontrollable one, which will be discussed in the second subsection. The last subsection will address the external child survival effect.

³The existence and properties of a corner solution are presented in the Appendix, as it may be of interest regarding potential unified growth dynamics.

3.1 Labour Supply and Fertility Decisions

There are two ways to explain the positive gradient between hours worked and fertility rates. The first one uses the budget constraint. Dividing both sides of equation (2) by the hourly wage $w_t h_t$ reveals that the labour supply is governed by the spending shares on consumption by parents and children:

$$n_t^* = \frac{c_t^*}{w_t h_t} + \frac{b_t^* z_t^*}{w_t h_t}. \quad (15)$$

Given that every individual in the household consumes a constant fraction of the hourly wage (see equations (10) and (11)), it readily follows that the household's total consumption depends on its size, which is determined by the fertility rate b_t^* . This means that parents with more children work more hours because they spend a larger fraction of their hourly wage on children's consumption.

An alternative intuition behind the positive gradient between fertility and labour supply decisions can be obtained by using the time restriction. First, one needs to decompose the labour supply into working hours to finance consumption by parents and children, denoted n_t^c and n_t^z respectively. The time constraint is then decomposed as follows

$$T = n_t^c + n_t^z + (\bar{\epsilon} + \epsilon_t) b_t. \quad (16)$$

This last equation shows that parents devote time to their children directly through childcare $(\bar{\epsilon} + \epsilon_t) b_t$, and indirectly through consumption n_t^z . As parents always work αT hours to finance their own consumption (see equation (10)), it is necessarily the case that they use the remaining $(1 - \alpha)T$ hours to satisfy their children's needs:

$$(1 - \alpha)T = n_t^z + (\bar{\epsilon} + \epsilon_t) b_t. \quad (17)$$

Equation (17) thus indicates that the labour supply moves in step with fertility because the composition of the time devoted to satisfying children's needs changes. When parents raise more children b_t , they need to work longer hours n_t^z in order to purchase more food, clothes, etc. As the number of children diminishes, the need for goods decreases, and parents can reallocate time from work to childcare $(\bar{\epsilon} + \epsilon_t) b_t$. The latter becomes clear when considering the spending rule (11), whereby the amount of time worked for children's consumption is given by $n_t^z = (\theta - \bar{\epsilon}) b_t$. Equation (17) can therefore be reformulated to obtain the trade-off between the quantity and quality of children in equation (13).

One important aspect about the labour-fertility gradient in equation (14) is that the ratio between working hours and fertility rates n_t^*/b_t^* increases when the number of children goes down as equation (18) points out:

$$\frac{n_t^*}{b_t^*} = \frac{\alpha T}{b_t^*} + (\theta - \bar{\epsilon}). \quad (18)$$

This is important because the empirical evidence indicates that the ratio n_t^*/b_t^* has indeed fallen over time across countries (see Figure 2).

3.2 The Internal Child Survival Effect

One of this model's main novelties is the existence of an internal child survival effect. It emerges because parents are aware that their children's health depends on nutrition, clothing, and other controllable (internal) factors encompassed in child consumption z_t^* . As it is to be expected, parents with higher hourly wages spend more income per child, and thus, they can expect more children to survive ($\partial s_t^*/\partial w_t h_t > 0$). In turn, greater chances of child survival encourage parents to educate each child more, causing changes in the optimal household allocations, which is what we call as the internal child survival effect.

Theorem 1. *Parents with higher hourly wages have fewer children, work fewer hours, and devote more time to children's education because their children have better survival chances.*

Proof: As parents with higher hourly wages spend more income per child (see equation (11)), it follows that $\partial \Phi_t^*/\partial w_t h_t < 0$. Applying the chain rule in equations (12), (13) and (14) therefore shows that greater chances of child survival lead to a lower fertility rate

$$\frac{\partial b_t^*}{\partial w_t h_t} = \frac{\partial b_t^*}{\partial \Phi_t^*} \frac{\partial \Phi_t^*}{\partial z_t^*} \frac{\partial z_t^*}{\partial w_t h_t} < 0,$$

more time devoted to children's education,

$$\frac{\partial \epsilon_t^*}{\partial w_t h_t} = \frac{\partial \epsilon_t^*}{\partial b_t^*} \frac{\partial b_t^*}{\partial w_t h_t} > 0,$$

and fewer hours worked

$$\frac{\partial n_t^*}{\partial w_t h_t} = \frac{\partial n_t^*}{\partial b_t^*} \frac{\partial b_t^*}{\partial w_t h_t} < 0. \quad \square$$

To understand the mechanism behind Theorem 1, it should be noted that equations (8) and (9) can be merged to obtain the optimality condition

$$\frac{\partial U}{\partial s} \frac{\partial s}{\partial z} \frac{z_t^*}{\theta - \bar{\epsilon}} = \frac{\partial U}{\partial h} \frac{\partial h}{\partial \epsilon}.$$

Following our assumptions about the child survival function and the education technology, this latter condition establishes that health and education investments go hand in hand. In other words, parents that spend more income per child will

also invest more time in their education, that is, $\partial \epsilon_t^* / \partial z_t^* > 0$. As parents with higher hourly wages spend more income per child and invest more time in their education, the cost of raising a child increases (see the right hand-side of eq. (7)), leading them to have fewer children. A smaller family involves a lower demand for goods, allowing parents to reallocate time from work to childcare.

The internal child survival effect is a novelty compared to the study by Strulik (2004b). It emerges because the controllable mortality factors can be improved by spending more income per child instead of time, causing transversal patterns in the use of time driven by differences in human capital (h_t). Interestingly, these transversal patterns are consistent with the empirical evidence (see Guryan et al. (2008), Bick et al. (2018) and Doepke et al. (2019)). A further novelty of this model is thus that economic growth, either in the form of technological progress (w_t) or human capital accumulation (h_t), leads to a reallocation of time from work to educational childcare that is consistent with the evidence (see Gimenez-Nadal and Sevilla (2012), Doepke et al. (2019) and Boppart and Krusell (2020)).

3.3 The External Child Survival Effect

In line with Strulik (2004b), this model predicts the existence of an external child survival effect. This notion refers to changes in the optimal household allocations caused by variations in the external mortality factors encompassed by the dynamic index X_t . Like the internal effect, improvements in uncontrollable mortality factors (higher X_t) increase child survival rates, changing the optimal household allocations as stated in Theorem 2.

Theorem 2. *When uncontrollable mortality factors improve, parents decide to have fewer children, devote less time to work, and invest more time in educating their children.*

Proof: Since improvements in uncontrollable child mortality lead to higher child survival rates, that is, $\partial \Phi_t^* / \partial X_t < 0$, applying the chain rule in equations (12), (13) and (14) implies that parents react by having fewer children

$$\frac{\partial b_t^*}{\partial X_t} = \frac{\partial b_t^*}{\partial \Phi_t^*} \frac{\partial \Phi_t^*}{\partial X_t} < 0,$$

by increasing the amount of time devoted to children's education,

$$\frac{\partial \epsilon_t^*}{\partial X_t} = \frac{\partial \epsilon_t^*}{\partial b_t^*} \frac{\partial b_t^*}{\partial X_t} > 0,$$

and by working fewer hours

$$\frac{\partial n_t^*}{\partial X_t} = \frac{\partial n_t^*}{\partial b_t^*} \frac{\partial b_t^*}{\partial X_t} < 0. \quad \square$$

The mechanism behind the external survival effect is the same as in the case of the internal one. Improved child survivability encourages parents to educate each child more, raising the marginal cost of having a child. Consequently, parents choose to have a smaller family, which therefore decreases their demand for goods and labour supply. In this case, what is actually new is that the labour supply does not remain constant, as it does in Strulik (2004b). Moreover, the magnitude of the external effect varies across individuals with their hourly wage. In particular, parents with higher hourly wages are less sensitive to improvements in the external mortality factors because their children already have better chances of survival. This is confirmed by the second cross-partial derivative in equation (12),

$$\frac{\partial^2 b_t^*}{\partial X_t \partial w_t h_t} = \frac{(1 - \alpha)T}{(\theta - \bar{\epsilon})} \frac{\partial^2 \Phi_t^*}{\partial X_t \partial w_t h_t} < 0.$$

From a macroeconomic perspective, this last result is important regarding the impact of health innovation and its diffusion. Through the external child survival effect (X_t), less developed countries may converge demographically with richer ones without achieving the same income level $w_t h_t$. In other words, two countries with different hourly wages can have similar fertility and child mortality rates because of the international diffusion of medical knowledge and public health policies. This may lead to demographic convergence despite persistent income differentials.

4 Conclusions

This paper shows that child survival might play a role in labour supply decisions if parents consider that spending more income per child improves their offspring's chances of survival. According to the model, both internal and external improvements in child survivability lead to a shift in the allocation of time from work to childcare that is consistent with the empirical evidence. For the sake of brevity, this analysis has been restricted to the effects upon the optimal household allocations (the demand side). However, these effects may also lead to interesting general equilibrium implications, with the following two highlights.

One of the model's predictions is that higher hourly wages discourage the labour supply of individuals. If the labour demand of firms also falls with the hourly wage, then the labour market could yield multiple equilibria. Two economies with the same fundamentals (preferences, technology, and initial conditions) may thus follow different equilibrium paths towards the same asymptotic steady state (local indeterminacy). In such case, despite the convergence in fertility, mortality and growth rates over the long-run, there would be permanent differences in income.

Another potentially fruitful avenue of research regards the impact of public healthcare upon the household allocations. Improvements in public healthcare programmes might prompt an external child survival effect resulting in a reallocation of time from work to childcare. In turn, labour income taxes have the exact opposite effect. It can therefore be expected that the efficiency with which it is spent determines the net effect of public healthcare programmes upon the allocation of time between work and childcare.

Appendix: Existence of a Corner Solution

This appendix discusses when does the model predict a corner solution without time investment in education, that is, $\epsilon_t^* > 0$. Such a corner solution may be of interest regarding unified growth dynamics because of its features on fertility and human capital accumulation, which may resemble the behaviour of a Malthusian economy. Equation (13) shows that the model's solution is interior if, and only if, the optimal fertility choice is below the threshold

$$b_t^* < \frac{(1 - \alpha)T}{\theta}.$$

This threshold depends on the total number of available hours T , so it could be interpreted as the biological maximum, which would ensure that the optimal solution is always interior. Suppose that this is not the case, and the biological maximum exceeds this threshold, allowing the existence of a corner solution. Replacing the optimal fertility choice (12) on the left hand-side of the above condition then yields a reformulated condition in terms of child survivability

$$\Phi_t^* < \frac{\theta - \bar{\epsilon}}{\theta}.$$

This condition indicates that the interior solution exists only when child survival rates are sufficiently high (Φ_t^* sufficiently low), which in turn depends on the economy's developmental stage captured by the hourly wage $w_t h_t$ and external mortality factors X_t . In short, the corner solution would take place in environments of low economic development (Malthusian regime).

The properties of the corner solution can be derived if $\epsilon_t^* = 0$ is set in the system of FOCs. The combination of equations (12) and (11)

$$\frac{1 - \Phi_t^*}{\Phi_t^*} z_t^* = w_t h_t \bar{\epsilon},$$

implies that child consumption z_t^* must be a strictly increasing and concave function of the hourly wage $w_t h_t$. Combining the first order condition of fertility, consumption and the budget constraint delivers the optimal fertility choice as a function of child consumption

$$b_t^* = \frac{(1 - \alpha)T}{\bar{\epsilon} + \frac{z_t^*}{w_t h_t}}.$$

Given the strict concavity of z_t^* with respect to $w_t h_t$, it follows that the optimal fertility rate b_t^* becomes an increasing function of the hourly wage $w_t h_t$ at the

corner solution. Finally, the optimal labour supply is computed using the budget constraint, whereby

$$n_t^* = \alpha T + (1 - \alpha)\Phi_t^* T.$$

Since Φ_t^* depends negatively on z_t^* , the labour supply decreases with higher hourly wages. In sum, the corner solution differs from the interior one in the way the internal child survival effect impacts the optimal fertility rate, turning the sign from negative to positive. Its effect upon the labour supply, however, remains the same. Regarding the external child survival effect, its impact becomes indeterminate because child consumption now depends on X_t .

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